
Deep Generative Models I

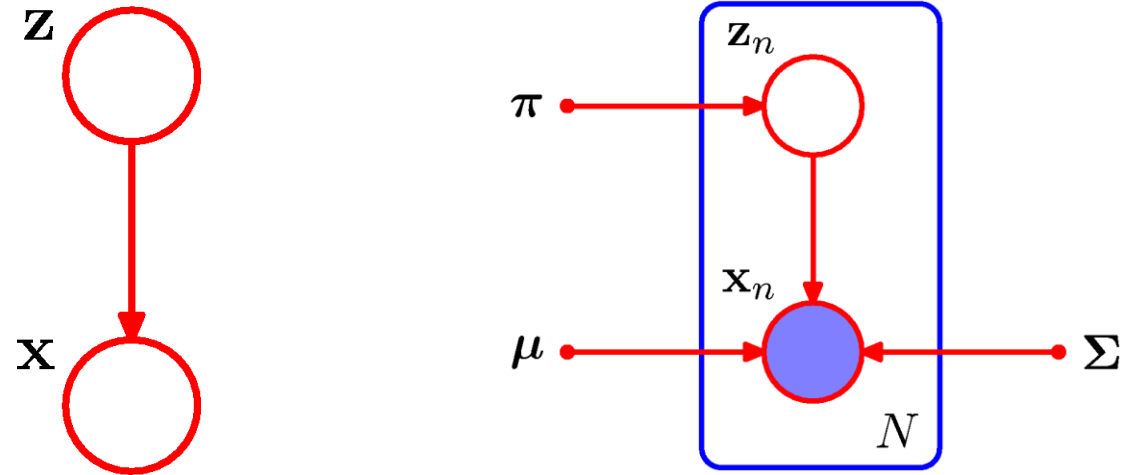
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Outline

- Auto-encoding Variational Bayes
- Generative Adversarial Networks

GMM with Latent Variables



Variational Inference

Latent variable Z 를 가진 모델의 데이터 x 가 주어졌을 때, marginal distribution은 다음과 같이 주어진다:

$$p(\mathbf{X}|\theta) = \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta)$$

- Discrete latent variable의 경우는 위와 같이 sum으로 표현
- Continuous latent variable의 경우 integral

Posterior $p(z|x)$ 를 근사하기 위한 확률분포 $q(Z|\theta')$ 를 제안
이때 이 확률분포 $q(Z|\theta')$ 가 다음 조건을 만족하는 경우를 생각해보자

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q||p)$$

Variational Inference

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q||p)$$

Where

$$L(q, \theta) = \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\}$$
$$KL(q||p) = - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)}{q(\mathbf{Z})} \right\}$$

- 위의 두 식은 범함수 (functional)은 함수들의 집합을 정의역으로 갖는 함수이다.
(ex: entropy, KL-divergence, Fourier transform 등)
 - 그러나 L의 경우, 파라미터 θ 에 대해서는 실수를 입력으로 하여 실수를 반환하는 일반함수
 - $L(\theta; q)$ 로도 나타냄
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Variational Inference

Prior knowledge

- **Jensen's Inequality** (옌슨 부등식)

- 만약 $f(x)$ 가 **convex** 를 만족한다면 다음이 항상 성립한다.

$$E[f(x)] \geq f(E[x])$$

- **Gibb's Inequality** (깁스 부등식)

- 함수 q 와 p 에 대해 $KL[p||q] \geq 0$ 을 항상 만족한다.

- (참고) 만약 $p = q$ 이면 $KL[p||q] = 0$

- (참고) 만약 $p \neq q$ 이면 $KL[p||q] > 0$

- (증명) $KL[p||q] = \sum_i p_i \ln \frac{p_i}{q_i} = - \sum_i p_i \ln \frac{q_i}{p_i} \geq - \ln[\sum_i p_i \frac{q_i}{p_i}] = - \ln[\sum_i q_i] = 0$

Variational Inference

In the aspect of Lower Bound

$$\ln p(\mathbf{X}|\theta) = \ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta)$$

이제 새로운 확률밀도함수 $q(\mathbf{z})$ 를 도입하여, 다음과 같이 표현할 수 있다:

$$\ln \sum_{\mathbf{Z}} p(\mathbf{X}, \mathbf{Z}|\theta) = \ln \sum_{\mathbf{Z}} q(\mathbf{Z}) \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \geq \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} = L(q, \theta)$$

**Jensen's
Inequality**

이와 같이 얻은 $L(\theta; q)$ 는 log likelihood $\ln p(\mathbf{X}|\theta)$ 의 lower bound (**E**vidence **L**ower **B**ound, ELBO)가 된다

Variational Inference

In the aspect of Lower Bound

ELBO가 되는 이유를 알아보기 위해, log likelihood $\ln p(\mathbf{X}|\theta)$ 와 $L(\theta; q)$ 의 차이를 계산해보자:

$$\begin{aligned}\ln p(\mathbf{X}|\theta) - L(q, \theta) &= \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{X}, \mathbf{Z}|\theta)}{q(\mathbf{Z})} \right\} \\ &= \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)p(\mathbf{X}|\theta)}{q(\mathbf{Z})} \right\} \\ &= \ln p(\mathbf{X}|\theta) - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)}{q(\mathbf{Z})} \right\} - \ln p(\mathbf{X}|\theta) \sum_{\mathbf{Z}} q(\mathbf{Z}) \\ &= - \sum_{\mathbf{Z}} q(\mathbf{Z}) \ln \left\{ \frac{p(\mathbf{Z}|\mathbf{X}, \theta)}{q(\mathbf{Z})} \right\} = KL[q(\mathbf{Z}) \| p(\mathbf{Z}|\mathbf{X}, \theta)] = KL[q \| p]\end{aligned}$$

이항하여 정리하면

$$\ln p(\mathbf{X}|\theta) = L(q, \theta) + KL(q \| p) \geq L(\theta; q)$$

Gibb's
Inequality

Variational Inference

In the aspect of **Posterior Approximation**

$$\begin{aligned} KL(q(z)||p(z|x)) &= \int q(z) \ln \frac{q(z)}{p(z|x)} dz \\ &= \int q(z) \ln \frac{q(z)p(x)}{p(x|z)} dz \\ &= \int q(z) \ln p(x) dz + \int q(z) \ln \frac{q(z)}{p(x, z)} dz \\ &= \ln p(x) - L(\theta; q) \geq 0 \end{aligned}$$

**Gibb's
Inequality**

Variational Inference

$$\ln p(x) = L(\theta; q) + KL(q(z)||p(z|x, \theta))$$

- We want to find proper $q(z)$ and θ that maximize target distribution $\ln p(x)$
- Hard to find optimal solution simultaneously
- Use Expectation-Maximization (find $q(z)$ and θ iteratively)

We have two issues:

- 1) how to find $q(z)$ and θ simultaneously?
 - 2) find $q(z)$ in the form of more complex function (related to x)
-

Variational Inference

Evidence Lower Bound

$$\log p_{\theta}(x) = KL(q_{\phi}(z)||p_{\theta}(z|x)) + L(\theta, \phi)$$

$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

We want to maximize lower bound in order to find $p_{\theta}(x)$

In GMM case using Expectation-Maximization,
we simply choose $q_{\phi}(z)$ as $p_{\theta}(z|x)$

What if the posterior is intractable, thereby impossible to choose?

Variational Inference

Evidence Lower Bound

$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

$$\begin{aligned} L(\theta, \phi) &= \int q_{\phi}(z) \log \frac{p_{\theta}(x, z)}{q_{\phi}(z)} dz \\ &= \int q_{\phi}(z) \log \frac{p_{\theta}(z) p_{\theta}(x|z)}{q_{\phi}(z)} dz \\ &= \int q_{\phi}(z) \log \frac{p_{\theta}(z)}{q_{\phi}(z)} dz + \int q_{\phi}(z) \log p_{\theta}(x|z) dz \\ &= -KL(q_{\phi}(z) | p_{\theta}(z)) + E_{z \sim q_{\phi}}[\log p_{\theta}(x|z)] \end{aligned}$$

Variational Inference

Evidence Lower Bound

$$\begin{aligned} L(\theta, \phi) \\ = -KL(q_\phi(z)|p_\theta(z)) + E_{z \sim q_\phi}[\log p_\theta(x|z)] \end{aligned}$$

$$\log p_\theta(x) \geq L(\theta, \phi)$$

We have two issues:

- 1) how to find $q(z)$ (in our case ϕ) and θ simultaneously?

Variational Inference

Evidence Lower Bound

$$L(\theta, \phi)$$

$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

$$= -KL(q_{\phi}(z)|p_{\theta}(z)) + E_{z \sim q_{\phi}}[\log p_{\theta}(x|z)]$$

We have two issues:

- 1) how to find $q(z)$ (in our case ϕ) and θ simultaneously?
- To relax the first issue, We want to **differentiate** and optimize the lower bound w.r.t. both the variational parameters ϕ and generative parameters θ .

Variational Inference

Evidence Lower Bound

$$L(\theta, \phi)$$

$$= -KL(q_\phi(z)|p_\theta(z)) + E_{z \sim q_\phi}[\log p_\theta(x|z)]$$

$$\log p_\theta(x) \geq L(\theta, \phi)$$

We have two issues:

1) how to find $q(z)$ (in our case ϕ) and θ simultaneously?

- To relax the first issue, We want to **differentiate** and optimize the lower bound w.r.t. both the variational parameters ϕ and generative parameters θ .

Variational Inference

Evidence Lower Bound

$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

$$E_{z \sim q_{\phi}}[\log p_{\theta}(x|z)]$$

The usual (naïve) Monte Carlo gradient estimator for this type of problem is:

$$E_{z \sim q_{\phi}}[\log p_{\theta}(x|z)] \cong \frac{1}{N} \sum_n \log p_{\theta}(x|z^n)$$

where

$$z^n \sim q_{\phi}(z)$$

Variational Inference

Evidence Lower Bound

$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

$$E_{z \sim q_{\phi}}[\log p_{\theta}(x|z)]$$

The usual (naïve) Monte Carlo gradient estimator for this type of problem is:

$$E_{z \sim q_{\phi}}[\log p_{\theta}(x|z)] \cong \frac{1}{N} \sum_n \log p_{\theta}(x|z^n)$$

where

$$z^n \sim q_{\phi}(z)$$

$$\begin{aligned} E_{z \sim q_{\phi}}[\log p_{\theta}(x|z)] \\ = \int q_{\phi}(z) \log p_{\theta}(x|z) dz \end{aligned}$$

$$\Rightarrow \nabla E_{z \sim q_{\phi}}[\log p_{\theta}(x|z)] \cong \frac{1}{N} \sum_n \nabla \log p_{\theta}(x|z^n)$$

Variational Inference

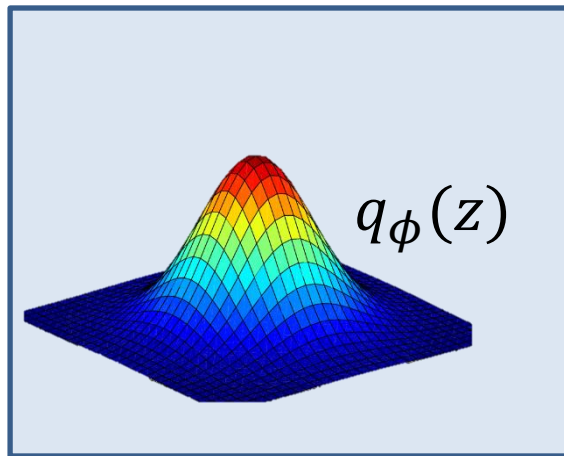
Evidence Lower Bound

$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

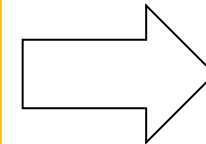
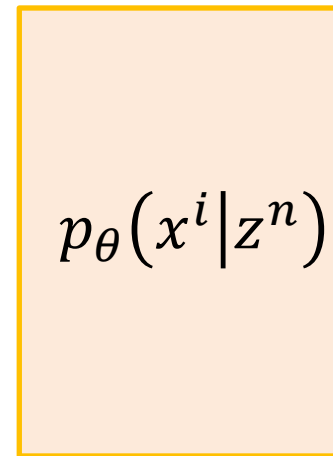
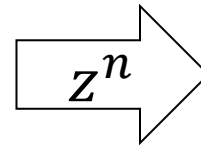
$$L(\theta, \phi) = -KL(q_{\phi}(z)|p_{\theta}(z)) + \frac{1}{N} \sum_n \log p_{\theta}(x|z^n)$$

where

$$z^n \sim q_{\phi}(z)$$



Sampling



x^i

Variational Inference

Evidence Lower Bound

$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

$$E_{z \sim q_{\phi}} [\log p_{\theta}(x|z)]$$

We have two issues:

2) find $q(z)$ in the form of more complex function (related to x)

For $q(z)$, we don't have any constraints of it; we can define $q(z)$ as we want:

$$q_{\phi} \triangleq q_{\phi}(z|x) \quad \text{Variational Likelihood}$$

$$\text{EX) } N\left(z; \mu_{\phi}(x^i), \sigma_{\phi}(x^i)^2 I\right)$$

Variational Inference

Evidence Lower Bound

$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

$$L(\theta, \phi)$$

$$= -KL(q_{\phi}(z|x)|p_{\theta}(z)) + \frac{1}{N} \sum_n \log p_{\theta}(x|z^n)$$

where

$$z^n \sim q_{\phi}(z|x)$$

Variational Inference

Evidence Lower Bound

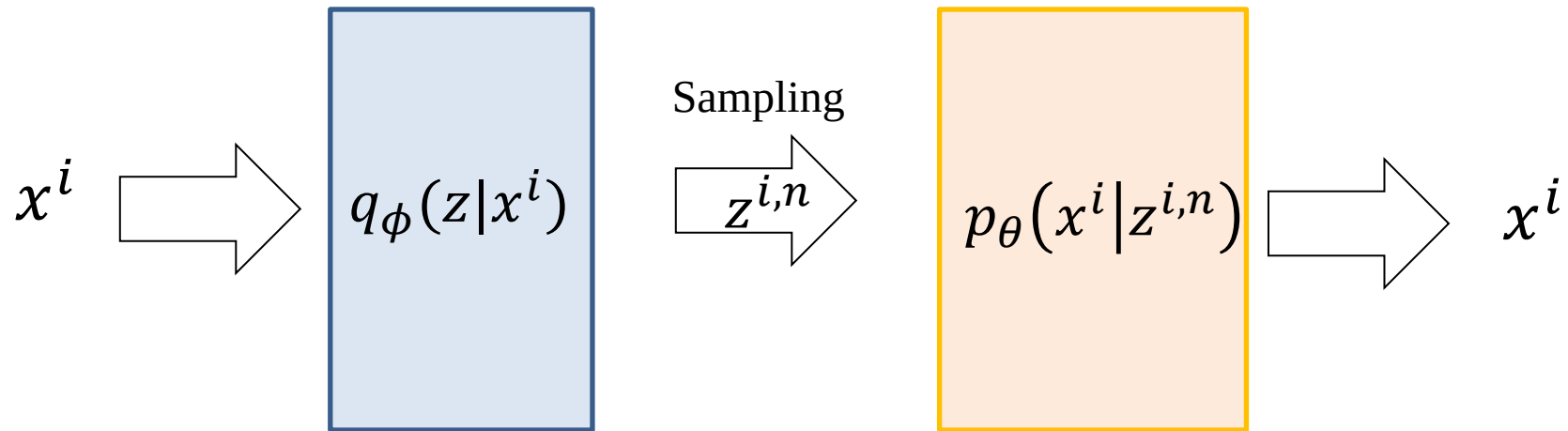
$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

$$L(\theta, \phi)$$

$$= -KL(q_{\phi}(z|x)|p_{\theta}(z)) + \frac{1}{N} \sum_n \log p_{\theta}(x|z^n)$$

where

$$z^n \sim q_{\phi}(z|x)$$



Variational Inference

Evidence Lower Bound

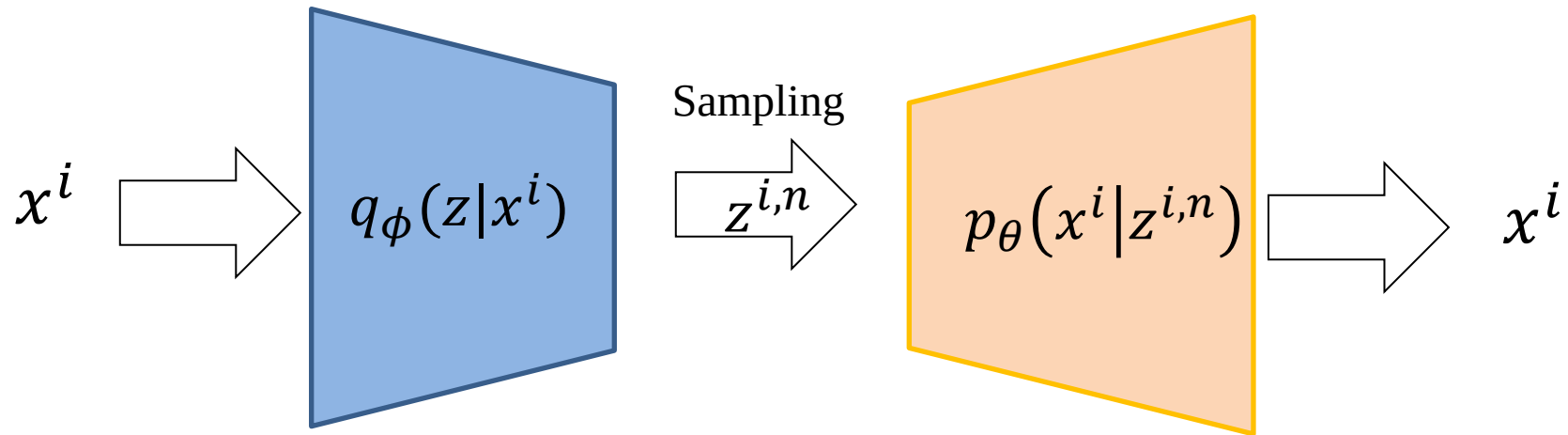
$$\log p_{\theta}(x) \geq L(\theta, \phi)$$

$$L(\theta, \phi)$$

$$= -KL(q_{\phi}(z|x)|p_{\theta}(z)) + \frac{1}{N} \sum_n \log p_{\theta}(x|z^n)$$

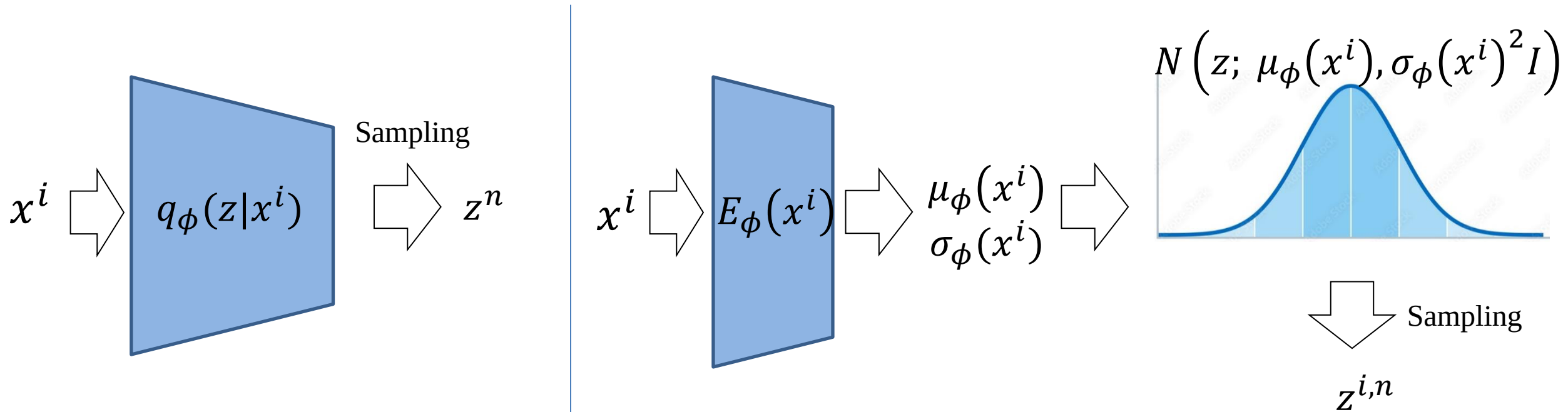
where

$$z^n \sim q_{\phi}(z|x)$$



Variational Inference

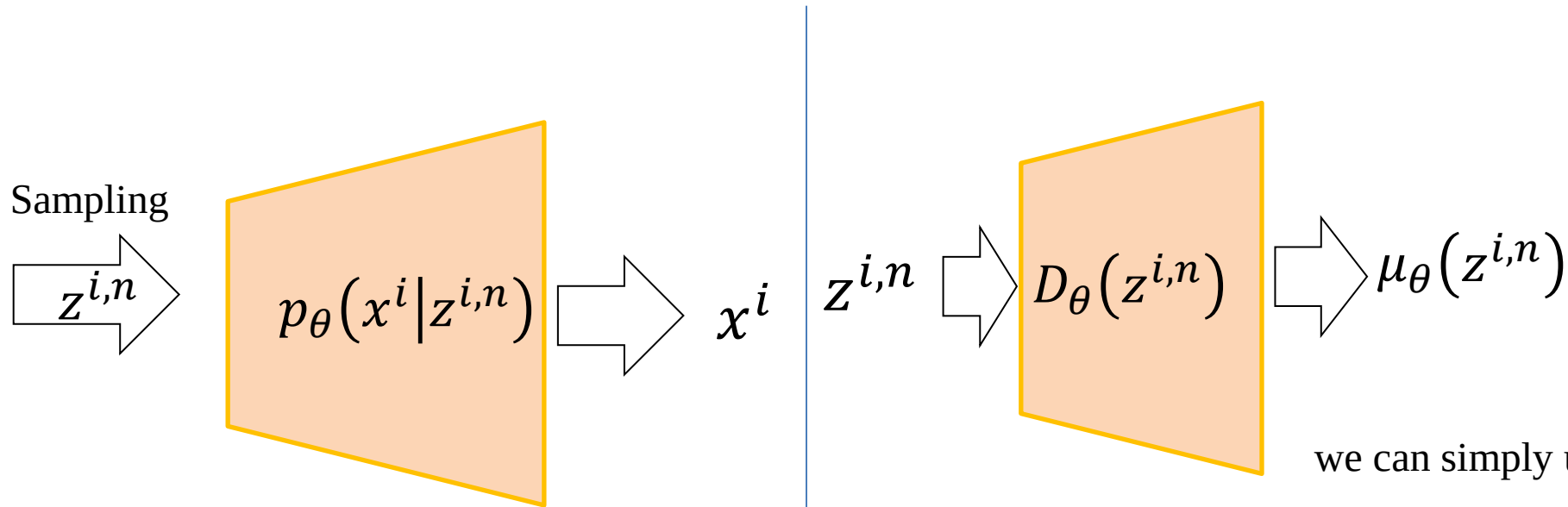
$$q_{\phi}(z|x) \triangleq N(z; \mu_{\phi}(x), \sigma_{\phi}(x)^2 I)$$



Variational Inference

Ex) image reconstruction

$$p_{\theta}(x^i | z^{i,n}) \triangleq N(x; \mu_{\theta}(z^{i,n}), \sigma^2 I)$$

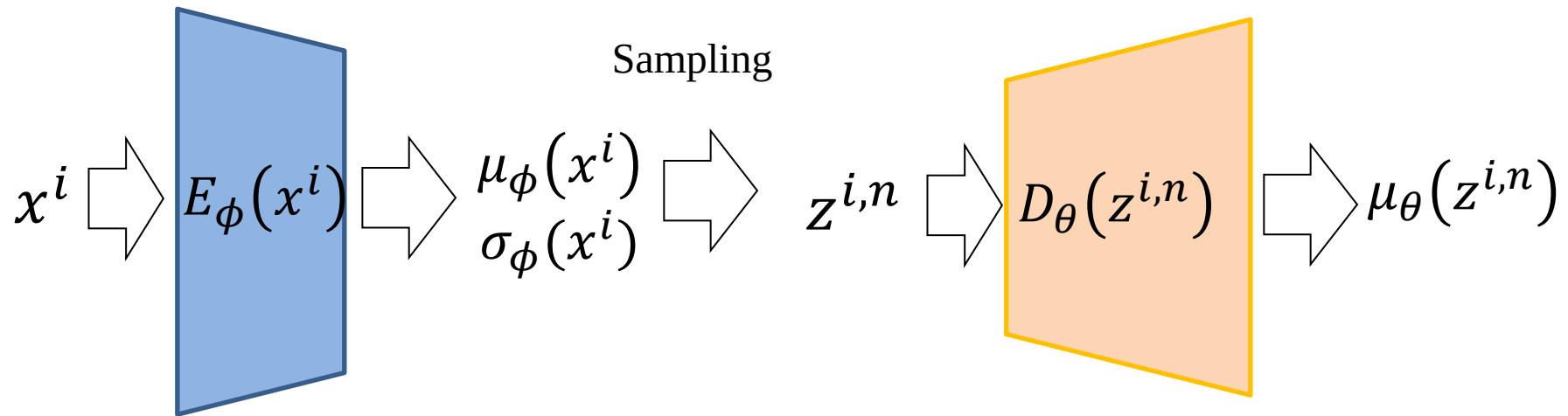


we can simply use MSE, by letting σ be a constant:

$$\begin{aligned} & \frac{1}{N} \sum_n |\mu_{\theta}(z^{i,n}) - x^i|^2 \\ &= \frac{1}{N} \sum_n |D_{\theta}(z^{i,n}) - x^i|^2 \end{aligned}$$

Variational Inference

$$L(\theta, \phi) = -KL(q_\phi(z|x)|p_\theta(z)) + \frac{1}{N} \sum_n \log p_\theta(x|z^n)$$

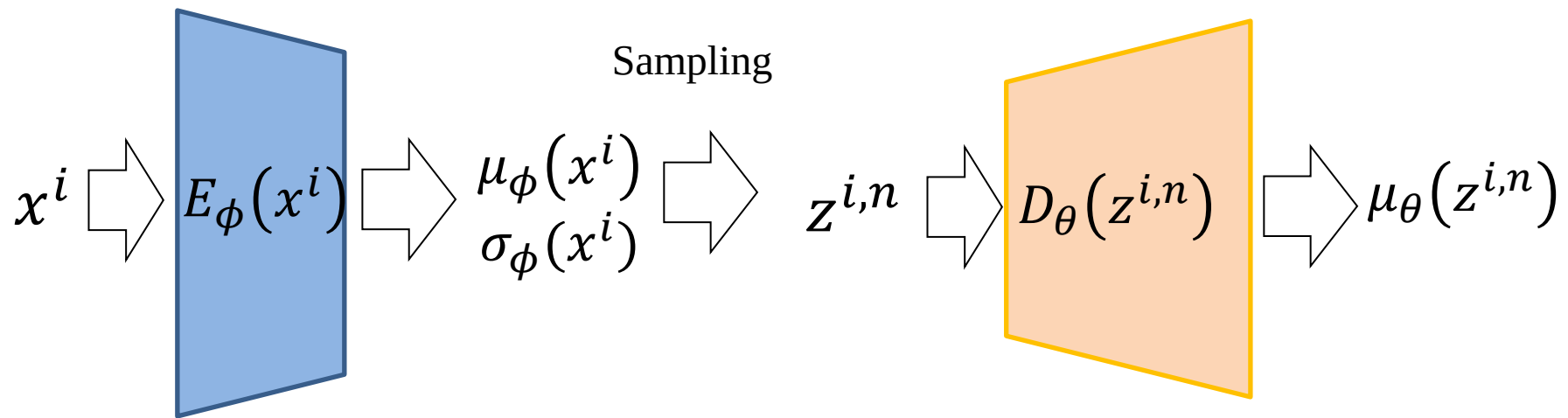


For loss, we can simply use MSE:

$$\frac{1}{N} \sum_n |\mu_\theta(z^{i,n}) - x^i|^2 = \frac{1}{N} \sum_n |D_\theta(z^{i,n}) - x^i|^2$$

Variational Inference

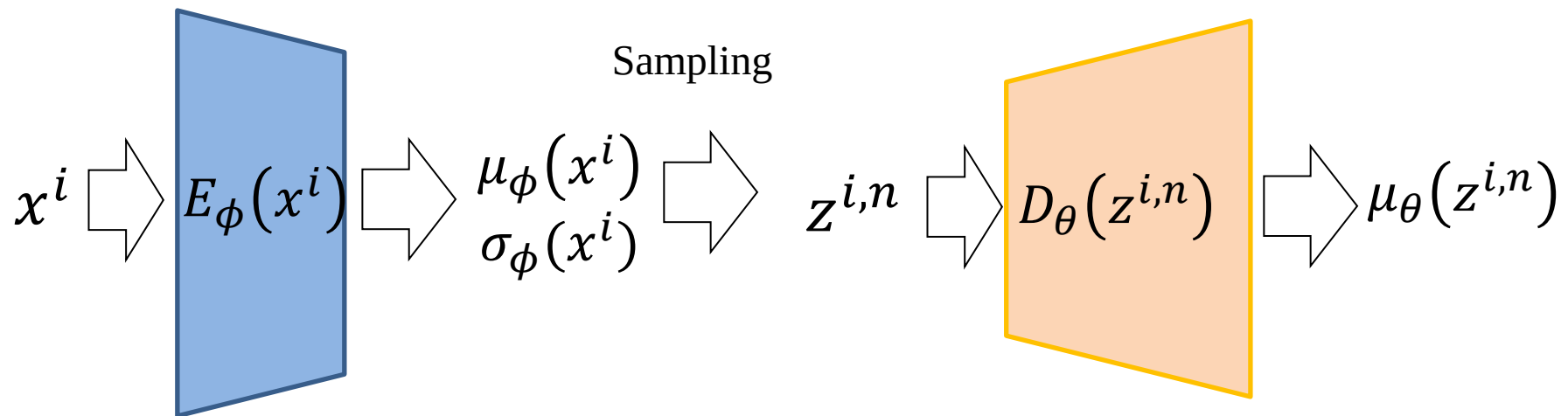
$$L(\theta, \phi) = \boxed{-KL(q_\phi(z|x)|p_\theta(z))} + \frac{1}{N} \sum_n \log p_\theta(x|z^n)$$



Variational Inference

$$L(\theta, \phi) = \boxed{-KL(q_\phi(z|x)|p_\theta(z))} + \frac{1}{N} \sum_n \log p_\theta(x|z^n)$$

Constraints for
latent variables



Variational Inference

$$L(\theta, \phi) = \boxed{-KL(q_\phi(z|x)|p_\theta(z))} + \frac{1}{N} \sum_n \log p_\theta(x|z^n)$$

Constraints for
latent variables

$$q_\phi(z|x) \triangleq N(z; \mu_\phi(x), \sigma_\phi(x)^2 I)$$

$$p_\theta(z) \triangleq N(z; 0, I)$$

Variational Inference

$$L(\theta, \phi) = \boxed{-KL(q_\phi(z|x)|p_\theta(z))} + \frac{1}{N} \sum_n \log p_\theta(x|z^n)$$

$$\int q_\theta(\mathbf{z}) \log p(\mathbf{z}) d\mathbf{z} = \int \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\sigma}^2) \log \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I}) d\mathbf{z}$$

$$= -\frac{J}{2} \log(2\pi) - \frac{1}{2} \sum_{j=1}^J (\mu_j^2 + \sigma_j^2)$$

$$\int q_\theta(\mathbf{z}) \log q_\theta(\mathbf{z}) d\mathbf{z} = \int \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\sigma}^2) \log \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}, \boldsymbol{\sigma}^2) d\mathbf{z}$$

$$= -\frac{J}{2} \log(2\pi) - \frac{1}{2} \sum_{j=1}^J (1 + \log \sigma_j^2)$$

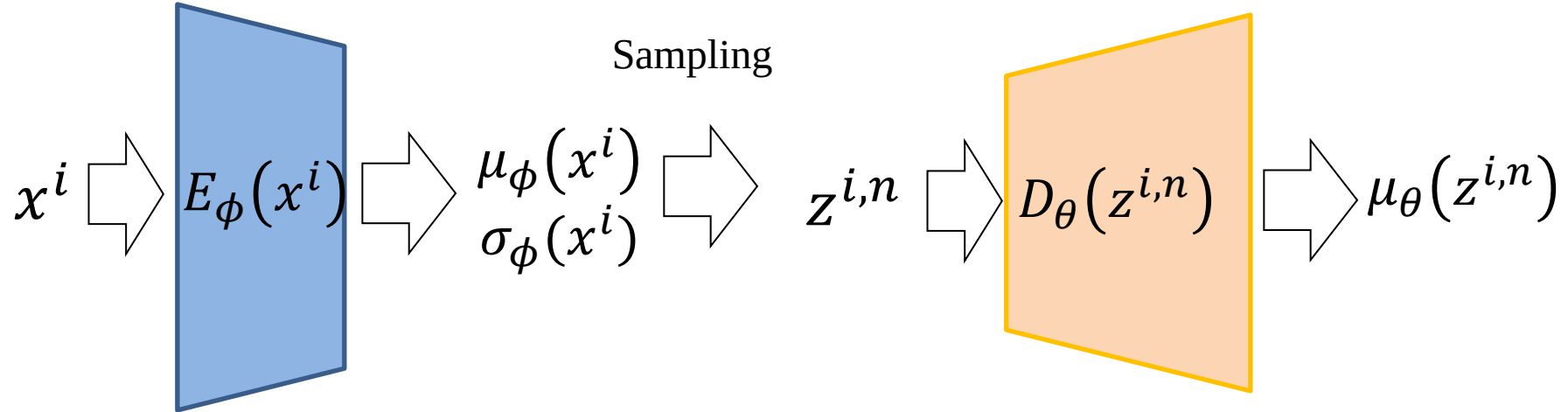
Variational Inference

$$L(\theta, \phi) = \boxed{-KL(q_\phi(z|x)|p_\theta(z))} + \frac{1}{N} \sum_n \log p_\theta(x|z^n)$$

$$\begin{aligned} -KL(q_\phi(z|x)|p_\theta(z)) &= \int q_\theta(\mathbf{z}) (\log p_\theta(\mathbf{z}) - \log q_\theta(\mathbf{z})) d\mathbf{z} \\ &= \frac{1}{2} \sum_{j=1}^J (1 + \log((\sigma_j)^2) - (\mu_j)^2 - (\sigma_j)^2) \end{aligned}$$

Variational Inference

$$L(\theta, \phi) = -KL(q_\phi(z|x)|p_\theta(z)) + \frac{1}{N} \sum_n \log p_\theta(x|z^n)$$



For loss, we can simply use MSE + KL-constraint term:

$$\frac{1}{N} \sum_n |D_\theta(z^{i,n}) - x^i|^2 + \frac{1}{2} \sum_{j=1}^J (1 + \log((\sigma_j^2)) - (\mu_j)^2 - (\sigma_j^2))$$

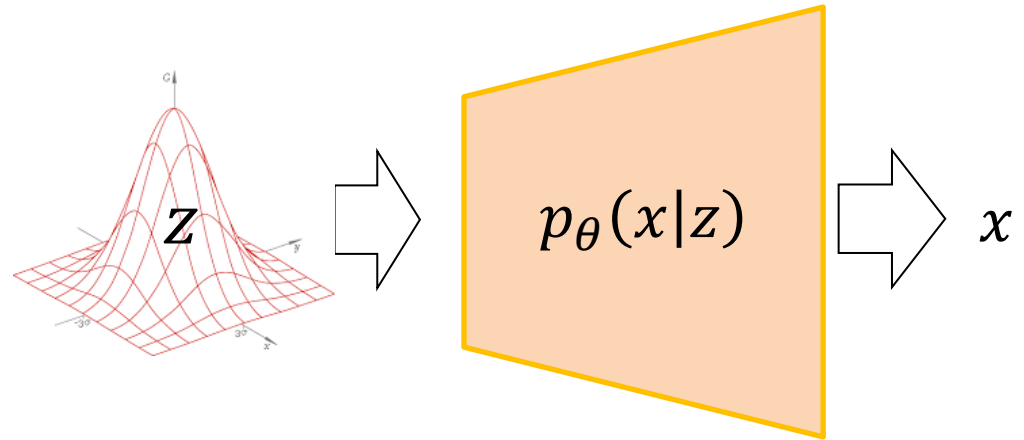
Variational Inference

Now we have:

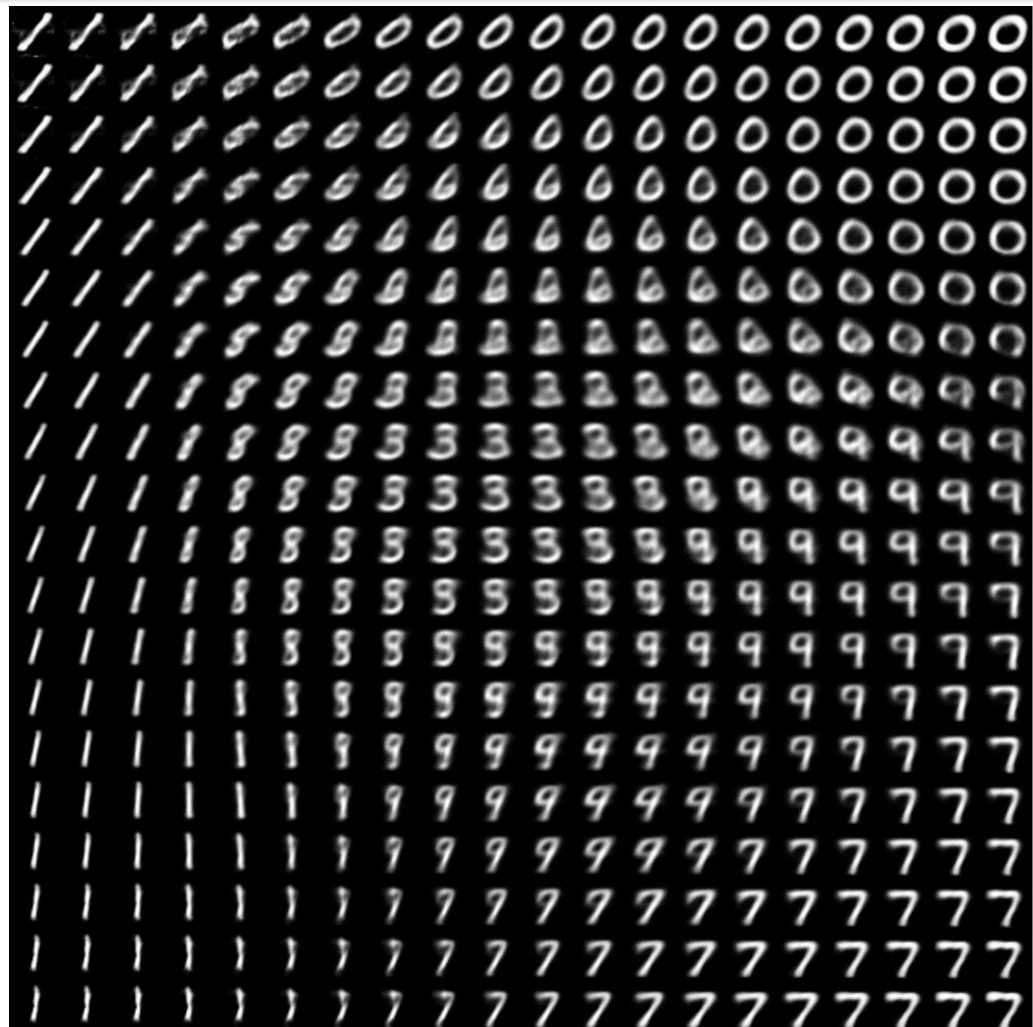
- 1) likelihood $p_\theta(x|z)$ (Decoder)
 - 2) posterior $p_\theta(z|x) \cong q_\phi(z|x)$ (Encoder)
 - 3) prior $p_\theta(z)$ (by assumption)
- Bayesian decision
 - Generative model

$$x \sim p_\theta(x) = \int p_\theta(x|z)p(z)dz$$

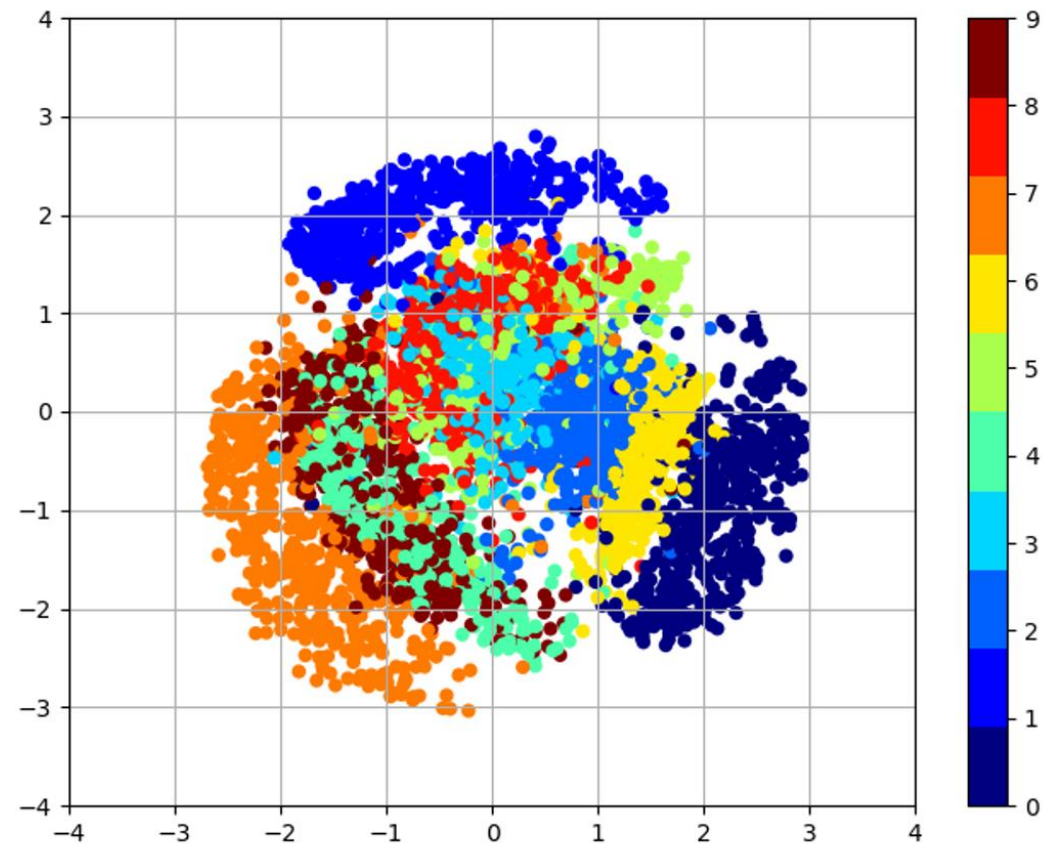
$$x \sim p_\theta(x|z) \text{ where } z \sim p_\theta(z)$$



Variational Inference



$$x \sim p_{\theta}(x|z) \text{ where } z \sim p_{\theta}(z)$$



$$z \sim q_{\phi}(z|x)$$

Interim Summary

- Auto-encoding Variational Bayes
- Generative Adversarial Networks